Xinghao Chen



Ph.D. Student in Computer Science
Research Interests:
Long-Context KV Cache Reuse
Latent-Space Reasoning
Thought Compression & Distillation

✉ xing-hao.chen@connect.polyu.hk
Google Scholar

Education

The Hong Kong Polytechnic University & Eastern Institute of Technology, Ningbo — Hong Kong / Ningbo, China
Joint Ph.D. Student in Computer Science — *Sep. 2024 – Present*

Supervisors: Wenjie Li, Professor Xiaoyu Shen, Assistant Professor

Nankai University, College of Artificial Intelligence — Tianjin, China
B.S. in Intelligent Science and Technology — *Sep. 2019 – Jun. 2023*

Internship Experience

Tencent Youtu Lab — Jun. 2026 – Present
Research Intern *Working with: Junnan Dong*

Publications

* indicates equal contribution, † indicates corresponding author.

Preprints

[1] X. Chen*, J. Dong*, C. Ke, C.T. Leong, H. Sun, K. Chen, S. An, R. Qiao, X. Sun, W. Li, X. Shen†. ATTUNER: Recomputation-Free Agentic Context KV Reuse. **Preprint**.

[2] C. Ke, X. Chen, X. Shen, K. Chen, S. An, J. Dong, R. Xu, R. Qiao, X. Sun. Disentangling Long-Term Memory via Latent Neuro-Symbolic Reasoning. **Preprint**.

[3] C. Ke, X. Chen, Y. Bei, Y. Lai, X. Shen, K. Chen, R. Qiao, S. An, J. Dong, R. Xu, X. Sun. Latent Multimodal Memory Reasoning. **Preprint**.

LLM Reasoning & Efficient Intelligence

[4] X. Chen, C.T. Leong, W. Guo, J. Wang, W. Li, X. Shen†. What Makes Effective Supervision in Latent Chain-of-Thought: An Information-Theoretic Analysis. [C]//**(ICML)** Forty-Third International Conference on Machine Learning. **CCF-A**
Established an information-theoretic framework dissecting latent CoT supervision into Trajectory Control and State Alignment; introduced the Unified Latent Probe (ULP) to reveal a strict information-performance binding between mutual information retention and reasoning accuracy.

[5] X. Chen*, A. Zhao*, H. Xia, X. Lu, H. Wang, Y. Chen, W. Zhang, J. Wang†, W. Li, X. Shen†. Reasoning Beyond Language: A Comprehensive Survey on Latent Chain-of-Thought Reasoning.[C]// **(EMNLP)** Findings of the 2026 Conference on Empirical Methods in Natural Language Processing. **CCF-B**
The *first* comprehensive survey on latent CoT reasoning, categorizing methods along two axes: *Token-wise Horizontal Level* and *Layer-wise Vertical Level*; maintaining a continuously updated repository at Awesome-Latent-CoT.

[6] X. Chen, Z. Sun, W. Guo, M. Zhang, Y. Chen, Y. Sun, H. Su, Y. Pan, D. Klakow, W. Li†, X. Shen†. Unveiling the Key Factors for Distilling Chain-of-Thought Reasoning.[C]// **(ACL)** Findings of the 2025 Annual Meeting of the Association for Computational Linguistics. **CCF-A**
Conducted systematic study on *what makes effective Chain-of-Thought supervision for smaller models*, revealing that the conventional wisdom of "stronger teacher, better student" does not hold in reasoning distillation.

[7] S. Lyu*, X. Chen*, Z. Sun, T. Liu, D. Zhu, X. Shen. When Compression Helps and When It Hurts: Condition-Aware Analysis of Chain-of-Thought Distillation.[C]// **(EMNLP)** The 2026 Conference on Empirical Methods in Natural Language Processing. **CCF-B**

[8] H. Chen*, J. Lin*, X. Chen, Y. Fan, X. Jin, H. Su, J. Dong, J. Fu, X. Shen. Rethinking Visual Layer Selection in Multimodal LLMs.[C]// **(EMNLP)** The 2025 Conference on Empirical Methods in Natural Language Processing. **CCF-B, Oral**

[9] Y. Chen, D. Zhu, Y. Sun, X. Chen, W. Zhang, X. Shen. The Accuracy Paradox in RLHF: When Better Reward Models Don't Yield Better Language Models.[C]// **(EMNLP)** The 2024 Conference on Empirical Methods in Natural Language Processing. **CCF-B**

[10] X. Lu, S. Liu, B. Yin, Y. Li, X. Chen, H. Su, Y. Jin, W. Zeng, X. Shen. MultiConIR: Towards Multi-Condition Information Retrieval.[C]// **(EMNLP)** Findings of the 2025 Conference on Empirical Methods in Natural Language Processing. **CCF-B**

[11] H. Wu, X. Wang, J. Zhang, J. Tong, X. Chen, J. Lin, Y. Ma, X. Shen. UTPTrack: Towards Simple and Unified Token Pruning for Visual Tracking.[C]// **(CVPR)** The 2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition. **CCF-A**

[12] J. Yang, X. Chen, Y. Liu, Z. Sun, W. Guo, X. Shen. AutoDataset: A Lightweight System for Continuous Dataset Discovery and Search. Arxiv, 2603.07271.

Applied AI (Pre-doctoral)

[13] X. Zhang, X. Chen, Y. Ding, Y. Zhang, Z. Wang, J. Shi, N. Johansson, X. Huang. Smart Real-Time Evaluation of Tunnel Fire Risk and Evacuation Safety via Computer Vision. **Safety Science**, 2024. **CAS Q1, JCR Q1**

[14] Y. Ding, X. Chen, Y. Zhang, X. Huang. Smart Building Evacuation by Tracking Multi-Camera Network and Explainable Re-Identification Model. **Engineering Applications of Artificial Intelligence**, 2025. **CAS Q1, JCR Q1**

[15] Y. Ding, X. Chen, Z. Wang, Y. Zhang, X. Huang. Human Behaviour Detection Dataset (HBDset) Using Computer Vision for Evacuation Safety and Emergency Management. **Journal of Safety Science and Resilience**, 2024.

[16] X. Chen, G. Cheng, S. Liu, S. Meng, Y. Jiao, W. Zhang, J. Liang, W. Zhang. Probing 1D Convolutional Neural Network Adapted to Near-Infrared Spectroscopy for Efficient Classification of Mixed Fish. **Spectrochimica Acta Part A**, 2022. **CAS Q2, JCR Q1**

Research Projects

Position-Independent KV Cache Reuse for Long-Context Agents Jun. 2026 – Present
Ongoing research

Developed *Attuner* [1] for position-independent caching (PIC) of reusable skills, documents, and memories. Identified attention routing as a source of quality loss and trained low-rank query adapters through full-context distillation to read frozen artifact KV states without recomputation. Recovers near-full-prefill quality at near-PIC warm-cache time to first token (TTFT).

Future directions:

- **Quality:** Validate PIC with repeated artifact injection across multi-turn dialogues; address artifact discovery and selection under distracting context after improving evidence utilization.
- **Efficiency:** Measure and optimize cache construction, storage I/O, transfer, and assembly costs; characterize the quality–latency–memory Pareto frontier across cache budgets and reuse frequencies.
- **Goal:** Enable reusable agent context that matches full-prefill quality at PIC-level latency within practical storage and bandwidth budgets.

Latent Memory Reasoning for Long-Term Multimodal Agents Feb. 2026 – Present
Ongoing research

Contributed to neuro-symbolic approaches that organize long-term interaction histories in latent space. LGM [2] disentangles textual memories into sparse concepts and builds query-conditioned graphs to reason over explicit and implicit preferences. LMMR [3] aligns text states and visual patches in a shared sparse concept space, models cross-modal and temporal relations in dialogue-turn graphs, and injects graph readouts into continuous latent reasoning while retaining native visual tokens for fine-grained, temporally grounded responses.

Latent Chain-of-Thought Reasoning Jan. 2025 – Feb. 2026
Core research direction during Ph.D.

Investigating how to internalize explicit reasoning into continuous latent spaces for efficient and faithful LLM inference. Established an information-theoretic framework for analyzing supervision signals in latent CoT [4]; authored the first comprehensive survey of the field [5]; and empirically identified key factors governing CoT distillation from large to small models [6]. Ongoing work explores extending latent reasoning representations to serve as compressed, reusable memory for agentic systems.

Service

Teaching Assistant, The Hong Kong Polytechnic University

- COMP 3122: Information Systems Development — Spring 2026
- COMP 5241: Software Engineering and Development — Fall 2025
- COMP 1411: Introduction to Computer Systems — Spring 2025
- COMP 5511: Artificial Intelligence Concepts — Fall 2024

Invited Talks

Reasoning Beyond Language: A Comprehensive Survey on Latent CoT Reasoning
Invited talk at NICE and CIPS [Video] *Jun. 2025*

Skills

- **Programming Languages**: Python, MATLAB, C++
- **LLM Frameworks**: LLaMA-Factory, VERL, vLLM, SGLang
- **AI-Assisted Development**: Claude Code, Cursor, OpenAI Codex, OpenClaw
- **Languages**: Mandarin (Native), English (Fluent)

Self-Evaluation

- Endlessly curious about new ideas and technologies. Committed to research that matters beyond the paper — bridging the gap from insight to real-world impact. Collaborative, outgoing, and motivated by a simple standard: work that is technically sound, practically grounded, and genuinely worth sharing.